Aishwarya H

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LinkedIn | GitHub | Portfolio

Professional Summary

Backend Developer specializing in **Python**, **FastAPI**, **Flask**, and **SQL Server**, with hands-on experience building **REST APIs**, **RBAC systems**, and **AI-powered backend applications**. Experienced in **RAG pipelines**, **semantic retrieval**, **LLM integration**, and **computer vision**.

Professional Experience

Probytes Solutions February 2026 – September 2026
Software Developer Intern

Worker Management System | *ASP.NET Web Forms, VB.NET, SQL Server* May 2026 – June 2026

- Developed a role-based employee management system using ASP.NET Web Forms, VB.NET, and SQL Server, supporting **500+ employee records**.
- Implemented RBAC for Admin, HR, and Manager roles to enforce secure access to employee data and application features.

Roles & Responsibilities

- Contributed to the HR Management System using ASP.NET Web Forms, VB.NET, and SQL Server, supporting employee data management, role-based access, and application workflows.
- Developed and maintained modules for worker management, search and filtering, bulk data import, user access, company settings, and expiry tracking, while troubleshooting application issues.

Projects

PolicyRAG — Policy Q&A System | *Python, FastAPI, RAG, LLM, SQL* September 2026 – Present

- Built a permission-aware **RAG pipeline** with FastAPI for Q&A across **15+ policy documents**.
- Implemented **chunking, embeddings, semantic retrieval, and reranking**, achieving **90%+ retrieval precision**.
- Designed **citation-grounded responses** linking answers to source passages for traceability.
- Implemented **RBAC and document-level access control**, passing **100% of access-control tests**.

Palm Fruit Ripeness Detection | *Python, YOLOv8, Flask, OpenCV, MySQL* March 2026 – May 2026

- Developed a **YOLOv8** palm fruit ripeness detector achieving **90%+ precision**.
- Fine-tuned **YOLOv8** on **3,000+ labeled images** across **3 ripeness classes**.
- Built a **Flask** app for image/video inference, processing streams at up to **10 FPS**.
- Integrated **OpenCV and MySQL** for video processing and storing predicted Image and Video.

Technical Skills

Programming Languages: Python, SQL, VB.NET, HTML, CSS
Backend & Web: FastAPI, Flask, ASP.NET Web Forms, REST API Design, RBAC
AI & Generative AI: RAG, LLM Integration, Embeddings, Semantic Retrieval
Machine Learning & Deep Learning: YOLOv8, Object Detection, Computer Vision, OpenCV
Databases: SQL Server, MySQL
Tools: Git, GitHub, Visual Studio, VS Code.

Education

Shanmuganathan Engineering College Pudukottai, Tamil Nadu
Bachelor of Engineering in Computer Science, CGPA: 8.2 *2022 – 2026*

The Leader Matric Higher Secondary School Karaikudi, Tamil Nadu
Higher Secondary (HSC), CGPA: 8.3 *2020 – 2022*